

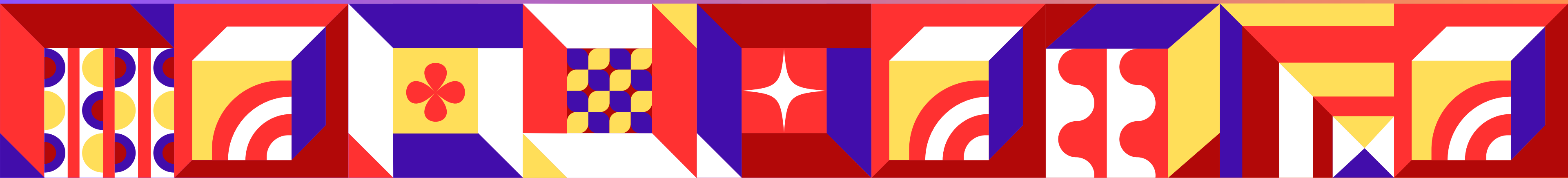


The Singular Haven for Harmony, Prose, and Theatrics!

MATINÉE

A curated collection of recommended books, movies, and
music.

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Project Overview

Objective

- Provides personalized recommendations for movies, music, and books using third-party APIs.
- Sleek UI with React and Tailwind CSS.
- Backend powered by Node.js, Express.js, and MongoDB for user preferences and interactions.

Scope and Limitations

- Features self and cross-category recommendations
- Does not implement existing API recommendations; uses its own logic for recommendations
- Movies ,Music and Books are limited to dataset



System Implementation

Design

- Tailwind CSS
- React with Vite
- Figma for background

Features

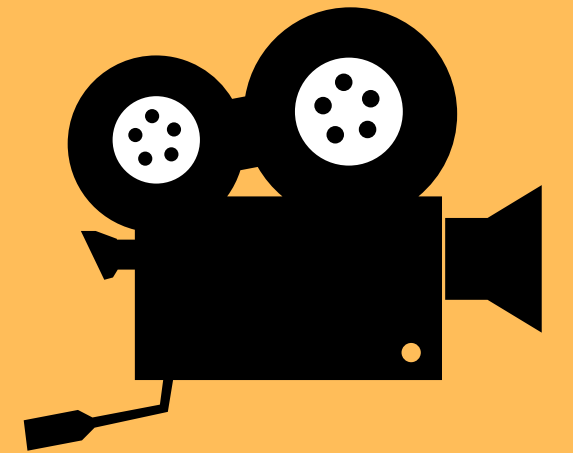
- Vectorized images
- Responsive buttons
- Navigation with react-router and react provider.

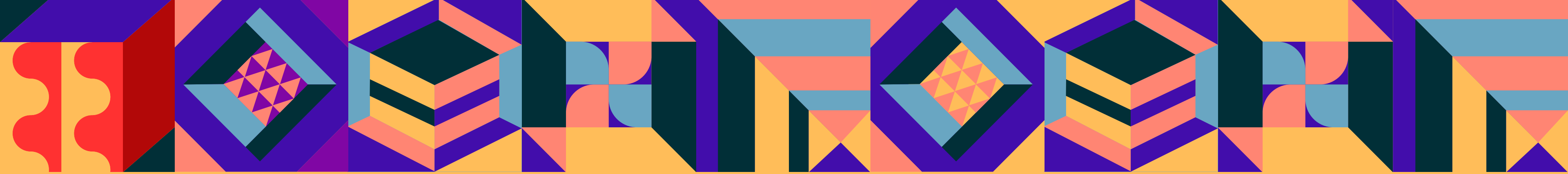
Components

- Moveable cards
- Footer with gradient color change
- About Us section
- Login feature
- Recommendation cards

Backend

- Connecting to MongoDB
- Data handling
- Text processing for vectorization
- Similarity calculation using TF-IDF and cosine similarity.
- Recommendation logic based on similarity scores.
- The scores are arranged in a descending order
- Authentication/User management with MongoDB integration.





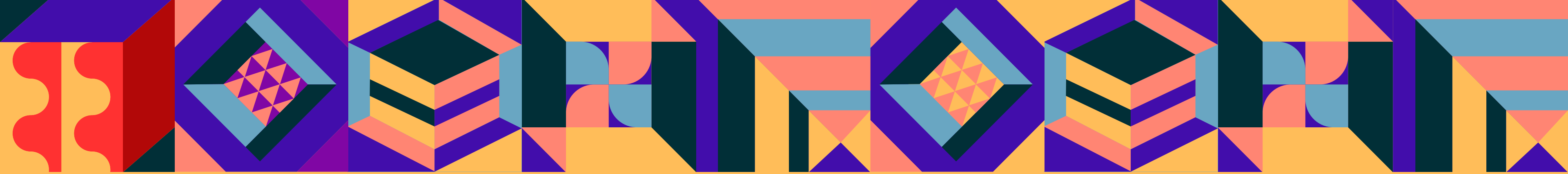
Code & Algorithm

Code snippets

- `server.js`
 1. Backend logic for data processing
 2. `vectorization`
 3. `recommendations.`
- `recommendation_logic.py`
- 4. Manages recommendation logic with similarity matrices.
- `index.js`
- 5. Manages user authentication, registration, & MongoDB connection for data storage.

Algorithms

- **TF-IDF Vectorization:** Converts combined text fields into numerical vectors using `TfidfVectorizer` from `sklearn`.
- **Cosine Similarity:** Computes similarity scores between vectorized items using `cosine_similarity` from `sklearn`.
- **Data Structures:**
- **DataFrames:** Used for movie, music, and book datasets.
- **Matrices:** TF-IDF feature matrices and cosine similarity matrices for similar results.



Conclusion And Future Work

Results

- Practical experience in full stack development.
- Successful implementation of personalized recommendations.
- Effective user authentication and management system.



Future Improvements

- Implement random recommendations for genre switching.
- Introduce mood-based recommendations to enhance user experience using Gen AI. Using NVIDIA Nim API

Thank You!